

Blood harmane is correlated with cerebellar metabolism in essential tremor: a pilot study.

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BACKGROUND: On proton magnetic resonance spectroscopic imaging ((1)H MRSI), there is a decrease in cerebellar N-acetylaspartate/total creatine (NAA/tCr) in essential tremor (ET), signifying cerebellar neuronal dysfunction or degeneration. Harmane, which is present in the human diet, is a potent tremor-producing neurotoxin. Blood harmane concentrations seem to be elevated in ET. OBJECTIVES: To assess in patients with ET whether blood harmane concentration is correlated with cerebellar NAA/tCr, a neuroimaging measure of neuronal dysfunction or degeneration. METHODS: Twelve patients with ET underwent (1)H MRSI. The major neuroanatomic structure of interest was the cerebellar cortex. Secondary regions were the central cerebellar white matter, cerebellar vermis, thalamus, and basal ganglia. Blood concentrations of harmane and another neurotoxin, lead, were also assessed. RESULTS: Mean +/- SD cerebellar NAA/tCR was 1.52 +/- 0.41. In a linear regression model that adjusted for age and gender, log blood harmane concentration was a predictor of cerebellar NAA/tCR (beta = -0.41, p = 0.009); every 1 g(-10)/mL unit increase in log blood harmane concentration was associated with a 0.41 unit decrease in cerebellar NAA/tCR. The association between blood harmane concentration and brain NAA/tCR only occurred in the cerebellar cortex; it was not observed in secondary brain regions of interest. Furthermore, the association was specific to harmane and not another neurotoxin, lead. CONCLUSION: This study provides additional support for the emerging link between harmane, a neurotoxin, and ET. Further studies are warranted to address whether cerebellar harmane concentrations are associated with cerebellar pathology in postmortem studies of the ET brain.